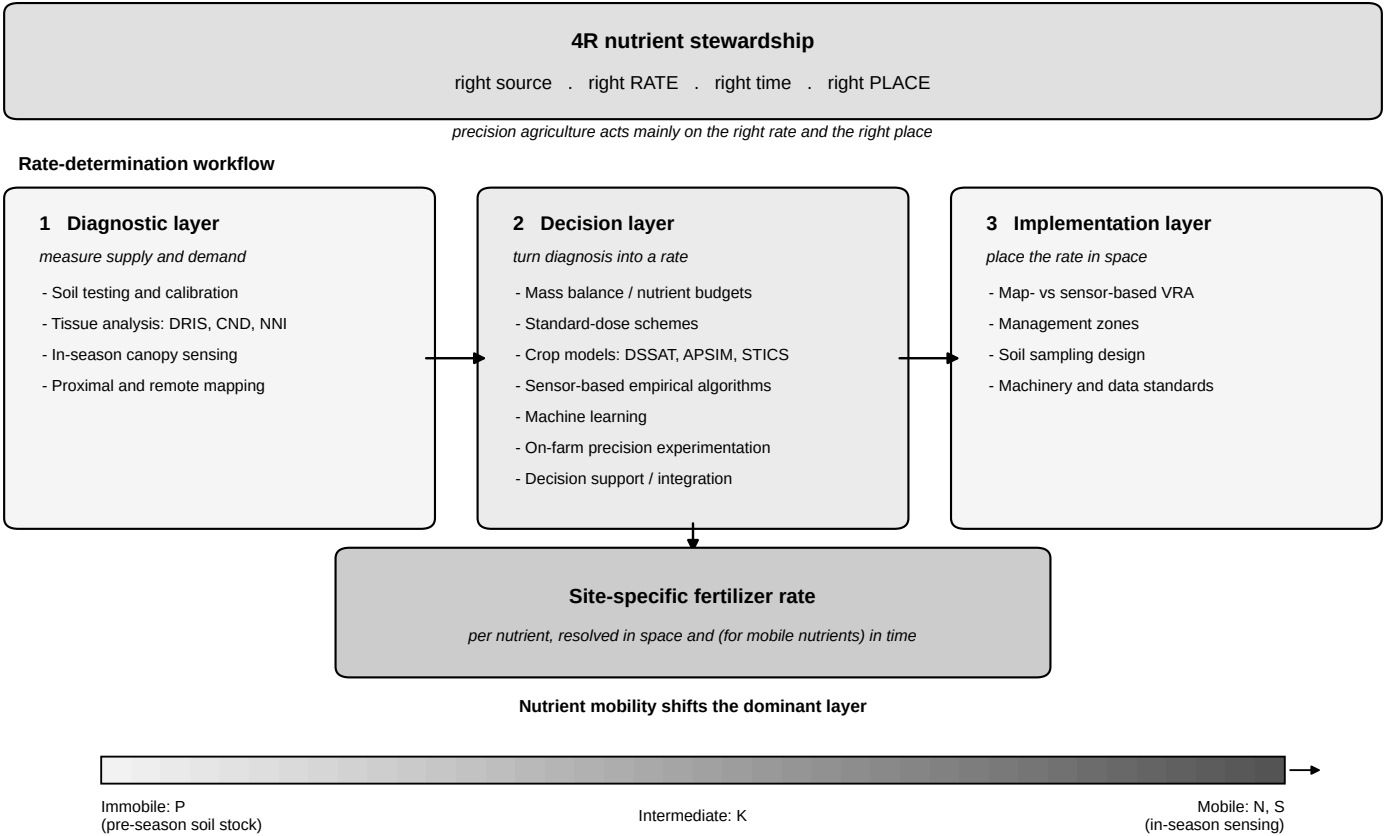
4R nutrient stewardship
right source . right RATE . right time . right PLACE
precision agriculture acts mainly on the right rate and the right place
Rate-determination workflow
1 Diagnostic layer
measure supply and demand
- Soil testing and calibration
- Tissue analysis: DRIS, CND, NNI
- In-season canopy sensing
- Proximal and remote mapping
2 Decision layer
turn diagnosis into a rate
- Mass balance / nutrient budgets
- Standard-dose schemes
- Crop models: DSSAT, APSIM, STICS
- Sensor-based empirical algorithms
- Machine learning
- On-farm precision experimentation
- Decision support / integration
3 Implementation layer
place the rate in space
- Map- vs sensor-based VRA
- Management zones
- Soil sampling design
- Machinery and data standards
Site-specific fertilizer rate
per nutrient, resolved in space and (for mobile nutrients) in time
Nutrient mobility shifts the dominant layer
Immobile: P
(pre-season soil stock)
Intermediate: K
Mobile: N, S
(in-season sensing)